

Solution Requirements

Date	13 July 2026
Team ID	
Project Name	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the Crop Recommendation System through the web application.	Medium
2	Authorization levels	General users (farmers) can enter crop parameters and receive recommendations. No separate admin role is implemented.	Low
3	External interfaces	<i>The system interacts with the trained Machine Learning model and crop dataset to generate recommendations through the Flask web interface.</i>	High

4	Transactions processing	<i>The system receives soil parameters (N, P, K, Temperature, Humidity, pH, Rainfall), validates them, sends them to the trained ML model, and generates a crop recommendation in real time.</i>	<i>High</i>
5	Reporting	<i>Display the recommended crop to the user. The system presents prediction results through the web interface.</i>	<i>Medium</i>
6	Business rules, etc.	<i>All seven input parameters must be provided within valid ranges. The ML model predicts only from the crops included in the trained dataset. Invalid or missing inputs are rejected.</i>	<i>High</i>
7	Compliance to laws or regulations	<i>The application does not collect or store personal user information. It is intended for educational and research purposes and follows responsible handling of agricultural data.</i>	<i>Low</i>
8	Other	<i>The application provides a simple, responsive, and user-friendly interface to help farmers quickly obtain crop recommendations.</i>	<i>Medium</i>

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>The system should process user inputs and generate a crop recommendation quickly using the trained machine learning model.</i>	<i>Prediction generated within 2–3 seconds.</i>
2	Scalability	<i>The application should support multiple users accessing the web interface simultaneously without significant performance degradation.</i>	<i>Supports at least 100 concurrent users.</i>
3	Security & Data Privacy	<i>[Detail data protection, encryption, and vulnerability management]</i>	<i>(e.g., End-to-end AES-256 encryption)</i>
4	Reliability & Availability	<i>The application does not store personal user information. User input is processed only for prediction, and server-side validation is performed.</i>	<i>No persistent storage of personal data; input validation for all fields.</i>
5	Usability & Accessibility	<i>The web interface should be simple, responsive, and easy to use on desktop and mobile devices for farmers and general users.</i>	<i>Responsive UI with clear input forms and readable prediction results.</i>
6	Other	<i>The application should have modular Flask code and a reusable ML prediction module so future crop datasets or models can be integrated easily.</i>	<i>Modular code structure that supports future model updates with minimal code changes.</i>